

Riley Beenders

OT NETWORKS | INDUSTRIAL AUTOMATION & CONTROLS | MULTIDISCIPLINE TECHNICAL PROJECTS

Bethlehem, PA | [\(484\) 523-4431](tel:(484)523-4431) | rileybeenders@gmail.com | www.rileybeenders.com | linkedin.com/in/riley-beenders

PROFESSIONAL SUMMARY

Engineering leader with 10+ years of multidisciplinary experience spanning industrial automation, manufacturing controls, and network infrastructure. Builds networked control systems with VLAN segmentation, firewall routing, and documented risk analysis, and leads multidiscipline technical projects from design standards and technical drawings through deployment, reliability, and support.

CORE EXPERTISE

Network & Security: Network infrastructure and design; firewall and reverse-proxy routing; server network and VLAN configuration; segmented industrial device networks; DNS and subdomains; domain-based application access; wireless deployments

Industrial Automation: Industrial automation and manufacturing controls; Modbus; CAN bus; Allen-Bradley PLCs; LabVIEW data acquisition; sensor and I/O integration; remote operations and diagnostics

Systems & Delivery: Windows/Linux systems administration; Proxmox, Docker, and Kubernetes; Python and PowerShell; security, operational, and software risk analysis; technical drawings and electrical schematics; technical writing; multidiscipline project leadership

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Performed risk analysis and implemented mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products, including networked remote operation, live video, and AI-assisted diagnostics.
- Built an automated LabVIEW QC system that negotiates Modbus communication with three tri-axial laser sensors over a dedicated private VLAN, charts live tolerances, and stores traceable reports on a remote server.
- Managed and expanded the division's IT infrastructure after the Proteor acquisition, deploying cloud gateways and standardizing software and internal processes to align with company-wide systems.
- Owned the mechanical, electrical, and software design of ICARUS-Lite through commercialization, including an IEEE-style electrical design adaptable to 120/240 VAC, and provide cross-functional technical support.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Progressed through four roles to CTO, overseeing the product-development pipeline, internal IT infrastructure, and technology strategy while staying hands-on across electrical, firmware, and software systems.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.
- Integrated a multi-board CAN-bus control architecture and independently testable electronics enclosures, simplifying wiring and debugging for larger, modular industrial machines.
- Expanded motion safety on ICARUS and POSEIDON with NEMA 34 motors, planetary gearboxes, and power-failure brakes, and led large automated systems through firmware tuning, operator training, and field support.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool.

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor and control safety checks on passenger lifts and supported maintenance during breakdowns while assisting thousands of guests.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: Lean Six Sigma Yellow Belt (2025); MathWorks MATLAB, Simulink, and Simscape Onramps (2026); Penn State Certificate in Engineering Design and Tools (2023)